

Variables and Constants Assignment

Flowchart

```
graph TD
    Start([Start]) --> V1[v]
    V1 --> Store10[Store 10 in var1]
    Store10 --> V2[v]
    V2 --> Store15[Store 15 in var2]
    Store15 --> V3[v]
    V3 --> Add[Add the two variables]
    Add --> V4[v]
    V4 --> Save[Save answer in result]
    Save --> V5[v]
    V5 --> End([End])
```

Challenges

A challenge I faced on this assignment was the concept of variables in Assembly language, as well as other programming languages. It also took a while to get the concept of initialized and uninitialized variables. Another difficulty was figuring out how to work the registers to add the values and store the correct total.

Assembly Code

```
section .text
    global _start

_start:
```

```
mov eax, [var1]
add eax, [var2]
mov [result], eax
```

```
mov eax, 1
int 0x80
```

```
segment .bss
result resb 4
```

```
segment .data
var1 dd 10
var2 dd 15
```

Result

10 + 15 = 25